retardants reduce the flame spread and rate of fire development by blocking the flames access to fuel and hindering future flame propagation. Boric acid/Borax, Di-ammonium Phosphate and Phosphoric acid, Sulfamic acid and Ammonium Sulfamate are a few substances used for non-durable flame retardant finishes. The durable flame retardant finishes include chemicals such as THPC- Tetakis Hydroxymethyl Phosphonium Chloride and its derivatives, N-Methyldimethyl Phosphonopropioamide, Phosphonic and Phosphoric Acid and its derivatives.



Figure 1.14 Flame Retardant Jacket

1.3.16 Anti-Microbial Finish

Antimicrobial finishes restrain the disease and decrease the risk of infection from following injury likes development of bacteria, other aroma causing germs, damage from perspiration and decay. These finishes are also called anti- bacteriostatic, germicidal or antiseptic finishes These finishes are mainly used for clothing that comes in close contact with the skin like shoe linings, hospital linings and carpeting. Wall covering and upholstery are also treated with antimicrobial chemicals. It is added to the spinning solutions in manufacturing fibres.

The most common chemical used for imparting anti-microbial finish is ziconium peroxide and sometimes an exposure to ethylene oxide gas is also used. Sutures, bandages and surgical gloves are treated with ethylene oxide because it is easy to available, low cost, safer and ultimate for medical products. The sterile environment to be maintained until the package is opened. Antimicrobial finish process includes gas treatment, chemical treatment and irradiation treatment.



Figure 1.15 Uses of Anti-Microbial Finished Fabrics

1.3.17 Antistatic Finish

Synthetic fibres of hydrophobic nature are prone to generation of static charges. This problem is very troublesome while processing the fabric at high speed in dry state. Static electricity is produced or created when two non-conducting surfaces such as synthetic textiles rub together. The two surfaces become oppositely charged and as the rubbing continues an electrical charge builds up. The wearer can experience the electric shocks and the fabric tends to cling to the body of the wearer. Antistatic finishes are chemical substances applied to reduce and eliminate static charge. The chemical substance used absorbs moisture from the atmosphere and thus reducing the

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